

Figure 1 illustrates the construction of a T -index for a graph. The figure consists of two rows of diagrams. The top row contains five diagrams, and the bottom row contains two larger diagrams. The diagrams use black, red, blue, and green edges and nodes to represent different components of the graph construction. The second diagram in the top row is labeled "Has a T -index". The second diagram in the bottom row is also labeled "Has a T -index".

EXAMPLE - 2

The diagrams illustrate various graph structures and their properties related to the T-index:

- Diagram 1 (Top Left): A graph with 4 vertices and 6 edges, labeled "Has a $T\backslash index$ ".
- Diagram 2 (Top Middle-Left): A graph with 4 vertices and 5 edges, labeled "Has a $T\backslash index$ ".
- Diagram 3 (Top Middle-Right): A graph with 4 vertices and 5 edges, labeled "Has a $T\backslash index$ ".
- Diagram 4 (Top Right): A graph with 4 vertices and 6 edges, labeled "Has a $T\backslash index$ ".
- Diagram 5 (Middle Left): A graph with 6 vertices and 10 edges, labeled "No $T\backslash index$ ".
- Diagram 6 (Middle Middle-Left): A graph with 6 vertices and 10 edges, labeled "No $T\backslash index$ ".
- Diagram 7 (Middle Middle-Right): A graph with 6 vertices and 10 edges, labeled "No $T\backslash index$ ".
- Diagram 8 (Middle Right): A graph with 6 vertices and 10 edges, labeled "No $T\backslash index$ ".
- Diagram 9 (Bottom Left): A graph with 6 vertices and 10 edges, labeled "No $T\backslash index$ ".
- Diagram 10 (Bottom Middle): A graph with 6 vertices and 10 edges, labeled "No $T\backslash index$ ".
- Diagram 11 (Bottom Right): A graph with 6 vertices and 10 edges, labeled "Has a $T\backslash index$ ".